



```

In [62]: #基本輸入輸出詳情範例參考MS2_compare_tool_v0
import pandas as pd
import re
file=pd.read_csv('zf_sPerMeNG_brain MS2 summary from 44218 at 20230307-153550.csv', sep='
print(file["peaklist"])
print(type(file["peaklist"][1]))

MSinput=[]
for i in range(len(file["peaklist"])):
    match = re.search(r'MS2peaklist¥(dMass=¥((.*?)¥)', file["peaklist"][i] )
    if match:
        # 取出所有浮點數
        ms2peaklist_str = match.group(1)
        #ms2peaklist = [round(float(x),2) for x in ms2peaklist_str.split(',') ]##
        ms2peaklist = [round(float(x),4) for x in ms2peaklist_str.split(',') ]
        #print(ms2peaklist)
        MSinput.append(ms2peaklist)
    else:
        print(' not match')
        break
print(len(MSinput))

def creat_Gdic(Gfrag, Gname):
    Gdic={}
    for i in range(len(Gname)):
        Gdic[Gname[i]]=Gfrag[i]

    return Gdic
def creat_Mdic(Gfrag, Gname):
    Mdic={}

    for n in range(len(Gfrag)):
        for t in Gfrag[n]:
            li=[]
            for i in range(len(Gfrag)):
                if t in Gfrag[i]:
                    #print(Gfrag[i])
                    li.append(Gname[i])
                    #print(li)
            Mdic[t]=li
    return Mdic

def creat_GFCount(gname, Gfrag):
    GFCount={}
    for i in range(len(gname)):
        GFCount[gname[i]]=len(Gfrag[i])
    return GFCount

```

```

0      MS2peaklist(dMass=(97.00896453857422, 98.33753...
1      MS2peaklist(dMass=(91.90139770507812, 91.91895...
2      MS2peaklist(dMass=(91.93927764892578, 94.68694...
3      MS2peaklist(dMass=(97.06316375732422, 97.59021...
4      MS2peaklist(dMass=(93.80081176757812, 98.26014...

...
23188  MS2peaklist(dMass=(92.02656555175781, 96.01469...
23189  MS2peaklist(dMass=(92.75369262695312, 92.77644...
23190  MS2peaklist(dMass=(90.8649673461914, 92.435340...
23191  MS2peaklist(dMass=(94.71047973632812, 97.10079...
23192  MS2peaklist(dMass=(95.1975326538086, 97.522605...
Name: peaklist, Length: 23193, dtype: object
<class 'str'>
23193

```

```

In [37]: #any(abs(2.02 - k) <= 0.2 for k in Mdic)
#讓2.05 in {2: ['G1', 'G2', 'G3'], 3: ['G2'], 4: ['G3']}=True 且等於 (返還) 2 如果差異太
def cusin(m, Mdic, tolerance):
    for k in Mdic:
        if abs(m - k) <= tolerance:
            return k
    return False
mdic={2: ['G1', 'G2', 'G3'], 3: ['G2'], 4: ['G3']}
cusin(2.05, mdic, 0.06)

```

Out[37]: 2

```

In [42]: ##原版
def creat_Td(M_list, Mdic):
    Td= {}
    for m in M_list:
        if m in Mdic:
            for T in Mdic[m]:
                if T in Td:
                    Td[T].append(m)
                else:
                    Td[T]=[m]
            else: continue
    return Td
creat_Td([0, 1, 2], {2: ['G1', 'G2', 'G3'], 3: ['G2'], 4: ['G3']})

```

Out[42]: {'G1': [2], 'G2': [2], 'G3': [2]}

```

In [97]: ##改版
def creat_Td_v2(M_list, Mdic, tolerance):
    Td = {}
    for m in M_list:
        if cusin(m, Mdic, tolerance):#
            m=cusin(m, Mdic, tolerance)#
            for T in Mdic[m]:#
                if T in Td:
                    Td[T].append(m)
                else:
                    Td[T]=[m]
            else: continue
    #print(Td)
    return Td
mdic={2: ['G1', 'G2', 'G3'], 3: ['G2'], 4: ['G3']}
#creat_Td_v2([1,2,3,4], {2: ['G1', 'G2', 'G3'], 3: ['G2'], 4: ['G3']}, 0.06)

##雖然在原本的m_list是2.05但會被視為與mdic中的2相等 並且在之後紀錄為2
creat_Td_v2([178.8598, 187.0959], {189.1121: ['F1H5N4S2', 'F1H6N4S2'], 205.1071: ['F1H5N

```

Out[97]: {'F1H5N4S2': [189.1121], 'F1H6N4S2': [189.1121]}

```

In [88]: """
Td={}
if cusin(187, {189.1121: ['F1H5N4S2', 'F1H6N4S2'], 205.1071: ['F1H5N4S2', 'F1H6N4S2']}, 10
    m=cusin(187, {189.1121: ['F1H5N4S2', 'F1H6N4S2'], 205.1071: ['F1H5N4S2', 'F1H6N4S2']})
    #print(m)
    for T in {189.1121: ['F1H5N4S2', 'F1H6N4S2'], 205.1071: ['F1H5N4S2', 'F1H6N4S2']}[m]
        print(T)
        if T in Td:
            Td[T].append(m)
        else:
            Td[T]=[m]
print(Td)
"""

```

```

F1H5N4S2
F1H6N4S2
{'F1H5N4S2': [189.1121], 'F1H6N4S2': [189.1121]}

```

```

In [119]: Gname=["F1H5N4S2", "F1H6N4S2"]
Gname=["F1H5N4S2", "F1H6N4S2", "F2H7N4S2", "F1H5N4S1", "F1H6N4S1_structure1", "F1H6N4S1_struct
F1H5N4S2=[189.1121, 205.1071, 246.1336, 264.1442, 344.1704, 376.1966, 450.2334, 452.249,
F1H6N4S2=[189.1121, 205.1071, 219.1227, 246.1336, 264.1442, 344.1704, 376.1966, 409.2068
F2H7N4S2=[189.1121, 205.1071, 219.1227, 246.1336, 278.1598, 344.1704, 376.1966, 409.2068
F1H5N4S1=[189.1121, 205.1071, 219.1227, 246.1336, 264.1442, 344.1704, 376.1966, 450.2334
F1H6N4S1_structure1=[189.1121, 205.1071, 219.1227, 246.1336, 264.1442, 344.1704, 376.196
F1H6N4S1_structure2=[189.1121, 205.1071, 219.1227, 246.1336, 278.1598, 344.1704, 376.196

Gfrag=[F1H5N4S2, F1H6N4S2, F2H7N4S2, F1H5N4S1, F1H6N4S1_structure1, F1H6N4S1_structure2]

```

```

Gname=["F1H5N4S2", "F1H6N4S2", "F2H7N4S2", "F1H5N4S1", "F1H6N4S1", "F2H6N4S1",
"F1H7N4S2", "F2H4N5", "F2H7N4S1G1", "F2H7N4G2", "F1H3N5"]

```

```
Gfrag=[[344.170, 376.197, 432.223, 450.233, 580.296, 793.397, 825.423], [344.170, 376.197, 432.223, 450.233, 580.296, 793.397, 825.423], [344.170, 376.197], [344.170, 376.197, 432.223, 450.233, 464.249, 580.296, 793.397, 825.423], [344.170, 376.197, 432.223, 450.233, 580.296, 793.397, 825.423], [344.170, 376.197, 432.223, 450.233, 464.249, 580.296, 793.397, 825.423], [344.170, 376.197, 432.223], [ 432.223, 464.249], [344.170, 374.181, 376.197, 406.207], [ 374.181, 406.207], [ 432.223, 464.249]]
```

```

In [120]: #替每個MSinput算出對應的Td_ratio並存在Tdlist

#Gfrag=[[2], [2, 3], [2, 4]]
#Gname=["F1H5N4S1", "G2", "G3"]
##MSinput=[[0.01, 1.03, 2.05], [1, 1.97], [1, 2, 3, 4], [3, 4, 5]]##
#MSinput=[[0, 1, 2], [1, 2], [1, 2, 3, 4], [3, 4, 5]]##
Gdic=creat_Gdic(Gfrag, Gname)
#print(' Gdic:', Gdic)
Mdic=creat_Mdic(Gfrag, Gname)
#print(' Mdic:', Mdic)
GFCount=creat_GFCount(Gname, Gfrag)
#print(' GFCount:', GFCount)

tolerance=0.1#

Tdlist=[]
c=0
for M_list in MSinput:
    ##Td=creat_Td(M_list, Mdic)##
    Td=creat_Td_v2(M_list, Mdic, tolerance)##
    Td_ratio={}
    for k, v in Td.items():
        #print(k, v)
        #print(' V:', v)
        Td_ratio[k] = len([m for m in v if m in Gdic[k]]) / GFCount[k]
    Tdlist.append(Td_ratio)
    #print(Td_ratio)
    """
    target="F1H5N4S2"
    threshold=0.6
    findMS=19462      #輸入MS2scanNumber 查找 不找為0
    #print(Td_ratio)

    if target in Td_ratio and Td_ratio[target]>threshold:
        print(' order:', c, 'MS2 Scan no', file['MS2 Scan no'][c], Td_ratio[target], file["in
    if findMS!=0 and file['MS2 Scan no'][c]==findMS:
        print(' -----')
        print(' order:', c, 'MS2 Scan no', file['MS2 Scan no'][c], Td_ratio)
        print(' -----')
    c+=1
    """

```

```

Out[120]: '   ¥n   target="F1H5N4S2"¥n   threshold=0.6¥n   findMS=19462      #輸入MS2scanNum
ber 查找 不找為0¥n   #print(Td_ratio)¥n¥n   if target in Td_ratio and Td_ratio[targ
et]>threshold:¥n           print('¥ order:¥', c, ¥' MS2 Scan no¥', file[¥' MS2 Scan no¥'] [c], T
d_ratio[target], file["in [H+]" ] [c])¥n   if findMS!=0 and file[¥' MS2 Scan no¥'] [c]==f
indMS:¥n           print('¥ -----¥')¥n           print('¥ order:¥', c, ¥' MS2 Scan
no¥', file[¥' MS2 Scan no¥'] [c], Td_ratio)¥n           print('¥ -----¥')¥n   c
+=1¥n'

```

```
In [121]: #len(Tdlist)#23193
#len(file)#23193
#len( MSinput)#23193
Tdlist[10103]
```

```
Out[121]: {'F1H5N4S2': 0.375,
'F1H6N4S2': 0.25396825396825395,
'F2H7N4S2': 0.1875,
'F1H5N4S1': 0.33962264150943394,
'F1H6N4S1_structure1': 0.2833333333333333,
'F1H6N4S1_structure2': 0.2}
```

```
In [122]: for c in range(len(file)):
            target="F1H5N4S2"
            threshold=0.62
            findMS=19462      #輸入MS2scanNumber 查找 不找為0
            #print(Td_ratio)

            if target in Tdlist[c] and Tdlist[c][target]>threshold:
                print('order:',c,'MS2 Scan no',file['MS2 Scan no'][c],Tdlist[c][target],file["in
            if findMS!=0 and file['MS2 Scan no'][c]==findMS:
                print('-----')
                print('order:',c,'MS2 Scan no',file['MS2 Scan no'][c],Tdlist[c])
                print('-----')
            c+=1
```

```
order: 9033 MS2 Scan no 17153 0.7 2569.307501152344
order: 9574 MS2 Scan no 18313 0.65 2569.3047545703125
order: 9781 MS2 Scan no 18726 0.625 2569.3025573046875
order: 9856 MS2 Scan no 18886 0.7 2773.405645683594
order: 9911 MS2 Scan no 19014 0.625 2379.2236998828125
order: 9943 MS2 Scan no 19085 0.65 2380.2256530078125
order: 9950 MS2 Scan no 19102 0.625 2773.4087584765625
order: 9965 MS2 Scan no 19141 0.675 2756.3896544726563
order: 9993 MS2 Scan no 19215 0.65 2379.2194884570317
```

```
-----
order: 10103 MS2 Scan no 19462 {'F1H5N4S2': 0.375, 'F1H6N4S2': 0.25396825396825395,
'F2H7N4S2': 0.1875, 'F1H5N4S1': 0.33962264150943394, 'F1H6N4S1_structure1': 0.28333
3333333333, 'F1H6N4S1_structure2': 0.2}
```

```
-----
order: 10391 MS2 Scan no 20070 0.7 2583.32074578125
order: 10396 MS2 Scan no 20084 0.7 2613.333196953125
order: 10404 MS2 Scan no 20106 0.775 2600.349005058594
order: 10423 MS2 Scan no 20145 0.625 2600.345431484375
order: 10426 MS2 Scan no 20148 0.625 2947.5047057421875
order: 10442 MS2 Scan no 20192 0.65 2920.4775450076562
```

```
In [102]: for i in range(len(file)):
            if abs(file["in [H+]" ][i]-2944)<2:
                print('order:',i,'MS2 Scan no',file['MS2 Scan no'][i],"H+":,file["in [H+]" ][i])
```

...

```
In [99]: def glycancomposition(a, b, c, d, e):  
        M = a * 174.08921 + b * 204.09977 + c * 245.12632 + d * 361.17367 + e * 391.18423  
        # M = a x Fucose (triangle) + b x Hexose (circle) + c x HexNAc (Square) + d * Neu5Ac  
        # a, b, c, d, e >=0, is integer  
        return(M)  
glycancomposition(1, 5, 4, 2, 0)
```

Out[99]: 2944.4898399999997

In [ ]:

In [ ]: